How Artificial Intelligence Is Changing Small and Medium-Sized Businesses in the UAE: Barriers to Effective Adoption and Practical Strategies

Abstract

Artificial intelligence (AI) is increasingly being incorporated into business activities such as marketing, customer service, analytics, forecasting, and administrative processes. Small and medium-sized enterprises (SMEs) are particularly important to the United Arab Emirates (UAE) economy, representing approximately 95% of businesses and around 86% of private-sector employment. At the same time, the UAE has developed a comparatively supportive environment for artificial intelligence through digital infrastructure, government initiatives, regulatory development, and investment in technological innovation.

However, access to AI technology does not necessarily result in effective adoption. SMEs may face organizational, technological, financial, skills-related, and governance challenges when attempting to integrate AI into their operations. This study examines the technological, organizational, and environmental factors influencing effective AI adoption among UAE SMEs and considers practical strategies for addressing implementation challenges.

The study uses a structured qualitative secondary research methodology, drawing primarily on UAE government publications, institutional research, and peer-reviewed academic literature. Two particularly relevant UAE sources provide empirical evidence: a 2025 Mohammed Bin Rashid School of Government study involving 81 UAE-based AI and digital SMEs, and a 2025 peer-reviewed study involving 315 respondents from UAE hospitality SMEs.

The evidence indicates that AI adoption is influenced by more than technological availability. Top-management support, competitive pressure, government regulation, perceived usefulness, competitive advantage, and employee capability are significantly associated with AI adoption intention in the studied UAE hospitality SME sample. UAE ecosystem research additionally highlights advanced computing requirements, financing, AI governance, talent, compliance, and scaling challenges. Broader SME literature reinforces the importance of skills, data readiness, implementation costs, uncertainty regarding returns, and organizational capability.

Based on this evidence, the study proposes an evidence-informed seven-phase framework: **Define, Assess, Select, Pilot, Measure, Govern, and Scale**. The framework is intended to help UAE SMEs move from initial AI experimentation toward adoption that is strategic, measurable, responsible, and scalable.

Keywords: Artificial intelligence; SMEs; UAE; AI adoption; digital transformation; Technology–Organization–Environment; responsible AI

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1. Introduction

Artificial intelligence is increasingly influencing how organizations perform tasks, analyze information, interact with customers, and make business decisions. Although large organizations often possess substantial financial and technical resources, SMEs frequently face greater constraints when adopting emerging technologies.

This makes SMEs an important group for AI research. AI may allow smaller firms to automate repetitive processes, improve responsiveness, strengthen decision-making, and access analytical capabilities that were previously more difficult or expensive to obtain. However, the benefits of AI are not automatic. International research indicates that SMEs can face barriers involving skills, data, financing, implementation costs, uncertainty about returns, organizational readiness, and legal or reputational risks.

The UAE provides a particularly relevant setting for examining these issues. According to the UAE Ministry of Economy, SMEs account for approximately 95% of businesses and around 86% of private-sector employment. This makes SME technological development relevant not only to individual firms but also to the broader UAE economy. (Ministry of Economy)

The UAE has also established a comparatively supportive AI environment. Research by the Mohammed Bin Rashid School of Government describes the UAE AI and digital SME ecosystem as supported by digital infrastructure, data governance, and regulatory development. The study is based on fieldwork involving 81 UAE-based AI and digital SMEs. (MBRSG)

However, a supportive national ecosystem does not mean that every SME can integrate AI effectively. The key issue is therefore not simply whether SMEs can access AI, but whether they possess the capabilities and conditions required to turn AI access into meaningful business value.

Research question

> **What technological, organizational, and environmental factors influence the effective adoption of artificial intelligence by SMEs in the UAE, and how can SMEs address the resulting implementation challenges?**

Research objectives

1. To identify the major technological, organizational, and environmental factors affecting AI adoption among UAE SMEs.
2. To examine the potential business value of AI for SMEs.
3. To analyze the challenges involved in moving from AI experimentation toward effective implementation.
4. To develop a practical, evidence-informed framework for responsible and scalable AI adoption.

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2. UAE SME and AI Context

SMEs constitute a substantial part of the UAE business environment. Their importance means that the ability of SMEs to adopt digital technologies can influence productivity, competitiveness, innovation, and economic diversification.

The UAE's AI ecosystem has developed alongside this SME environment. The MBRSG study of 81 UAE-based AI and digital SMEs describes the country as having comparatively strong digital foundations and an enabling ecosystem. It also indicates that the challenges faced by AI-oriented SMEs are increasingly associated with advanced capabilities rather than basic digital connectivity. (MBRSG)

This distinction is important.

An SME may have:

- reliable internet connectivity;
- access to cloud-based AI applications;
- generative-AI tools;
- digital payment systems;

while still lacking:

- advanced computing capability;
- suitable data;
- AI-skilled employees;
- management readiness;
- financial resources for scaling;
- governance capability.

The UAE therefore provides an environment in which the central question is increasingly **how effectively SMEs can use available AI capabilities**, rather than simply whether AI technology exists.

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3. Literature Review

3.1 Technological factors

Technology is an important component of AI adoption, but the literature suggests that technological readiness has several dimensions.

A 2025 systematic review of 78 peer-reviewed studies identifies technology readiness, customization, AI tools and needs, and data requirements among the major dimensions affecting SME AI adoption. It also identifies skills, financial readiness, management support, competitive pressure, partnerships, and regulatory compliance, demonstrating that technology cannot be considered independently of organizational and environmental conditions.

The OECD similarly identifies data, digital infrastructure, complementary investment, and implementation costs as important considerations for SMEs. It notes that cloud-based AI services can help SMEs access AI without building all capabilities internally, although issues such as data ownership, portability, and vendor lock-in can remain relevant.

The UAE evidence is consistent with this broader literature. The MBRSG study indicates that UAE AI and digital SMEs generally operate within a relatively strong digital environment but can encounter challenges involving access to advanced computing resources and other requirements for scaling AI capabilities. (MBRSG)

Therefore, technological readiness should be understood as **more than basic connectivity**. It includes whether an SME can access and integrate appropriate technology, data, computing resources, and supporting systems.

3.2 Organizational factors

Organizational readiness is one of the strongest themes in the literature.

Thomas et al. (2025) examined AI adoption intention among UAE hospitality SMEs using 315 respondents and a quantitative model combining the Technology–Organization–Environment framework with elements of the Technology Acceptance Model. The study found statistically significant relationships for six factors. Top-management support had the strongest reported coefficient ($\beta = .48$), followed by competitive pressure ($\beta = .47$), government regulation ($\beta = .43$), perceived usefulness ($\beta = .39$), competitive advantage ($\beta = .38$), and employee capability ($\beta = .30$). (SAGE Open)

These results suggest that AI adoption is not simply an IT decision.

Management influences:

- investment priorities;
- willingness to experiment;
- employee support;
- resource allocation;
- organizational change.

Employee capability is also important. Broader OECD research emphasizes that skills shortages can restrict SME AI adoption and that AI-related skills extend beyond advanced programming to digital, analytical, and data-related capabilities.

The implication is that SMEs should develop AI capability progressively rather than assuming every employee needs to become a technical AI specialist.

3.3 Financial readiness and uncertainty

Financial resources are another recurring issue.

The OECD identifies high implementation costs, complementary investment requirements, and uncertainty about AI returns as barriers for SMEs. It notes that SMEs can struggle to determine the business case for AI because productivity benefits may take time to appear and returns can be difficult to estimate.

The MBRSG research similarly identifies financing considerations within the UAE AI SME ecosystem, particularly in relation to growth and scaling. (MBRSG)

However, this study does not treat financing as the strongest empirically established determinant of AI adoption among all UAE SMEs. Instead, it is interpreted as an important **implementation and scaling condition**.

This distinction is important because saying that “UAE SMEs cannot afford AI” would be an unsupported generalization.

3.4 Environmental factors

The environment surrounding an SME can influence AI adoption through competition, government policy, regulation, partnerships, and external support.

The UAE hospitality SME study found competitive pressure to have a strong relationship with AI adoption intention ($\beta = .47$), while government regulation also showed a significant relationship ($\beta = .43$). (SAGE Open)

Competitive pressure can encourage firms to investigate AI because competitors may already be using it. However, competition should be treated as a **signal for evaluation rather than a reason for automatic adoption**.

Regulation also has a dual role. Clear regulation can support trust and responsible use, while compliance can require additional resources and expertise.

The UAE's AI policy framework emphasizes principles including accountability, transparency, explainability, safety, privacy, and human-centered values. These principles are relevant when SMEs move from experimentation toward real-world deployment.

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4. Methodology

4.1 Research design

This study uses a **structured qualitative secondary research methodology**.

The study does not claim to have conducted new surveys or interviews. Instead, it synthesizes existing evidence from:

- UAE government publications;
- UAE institutional reports;
- UAE-specific empirical studies;
- peer-reviewed academic research; and
- selected international literature on SME AI adoption.

4.2 Source-selection criteria

Sources were prioritized according to:

Geographic relevance: UAE evidence was prioritized.

Business relevance: Sources needed to address SMEs, AI adoption, entrepreneurship, digital transformation, or related business issues.

Credibility: Peer-reviewed studies, government publications, and established research organizations were prioritized.

Recency: Recent evidence was prioritized because AI technologies and adoption patterns are changing rapidly.

Methodological transparency: Empirical studies with identifiable samples and research methods were given greater weight.

4.3 Analytical approach

The Technology–Organization–Environment (TOE) framework was used as an **analytical lens**, rather than being statistically tested by this study.

Evidence was organized into:

Technology

- perceived usefulness;
- digital/technical readiness;
- data;

• advanced computing.

Organization

• management support;
• employee capability;
• financial readiness;
• organizational preparedness.

Environment

• competitive pressure;
• regulation;
• external ecosystem support.

The findings were then compared and synthesized to identify recurring patterns and develop an evidence-informed practical framework.

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5. Findings and Discussion

5.1 AI adoption should be distinguished from AI access

The evidence suggests that the UAE has established comparatively strong foundations for digital and AI development. However, access to AI tools alone does not guarantee successful integration.

An SME may experiment with generative AI for marketing or customer service without having the organizational systems required to integrate AI strategically into its operations.

Therefore, this study distinguishes between:

AI access — the ability to obtain or use an AI tool.

AI experimentation — trying AI for a limited task.

Effective AI adoption — integrating AI into a business process in a way that creates measurable value while remaining responsible and scalable.

The third concept is the focus of this research.

5.2 Leadership is central

The strongest reported coefficient in the UAE hospitality SME study was top-management support ($\beta = .48$). (SAGE Open)

This suggests that management commitment is an important condition for AI adoption intention.

The finding can be interpreted through the practical reality of SMEs: owners and managers often influence investment decisions, organizational priorities, employee training, and risk tolerance.

Therefore, AI should be treated as a **business transformation issue**, rather than simply an IT project.

5.3 Skills matter, but skills alone are insufficient

Employee capability showed a significant relationship with adoption intention ($\beta = .30$). (SAGE Open)

The broader literature also identifies skills as a major SME challenge. The OECD notes that AI adoption requires a combination of technical and domain expertise and that SMEs can face difficulties accessing AI talent.

However, the UAE evidence does not justify claiming that employee skills are the single biggest barrier.

The more defensible conclusion is:

> **AI capability must be developed alongside leadership commitment, business objectives, technology, and organizational readiness.**

5.4 SMEs need evidence of business value

Perceived usefulness had a significant relationship with AI adoption intention in the UAE study ($\beta = .39$). (SAGE Open)

International OECD evidence also highlights the difficulty businesses face in estimating AI return on investment.

This supports one of the central principles of this research:

> **SMEs should begin with a business problem rather than beginning with an AI tool.**

For example, instead of asking:

> “Should we implement an AI chatbot?”

an SME should ask:

> “Can AI reduce customer-response time without reducing service quality?”

The second question creates a measurable business case.

5.5 Competitive pressure is a driver, not proof of value

Competitive pressure had a strong relationship with adoption intention ($\beta = .47$). (SAGE Open)

This indicates that SMEs may be motivated to explore AI when competitors adopt it.

However, competitive pressure can also encourage poorly justified investments.

Therefore, this study recommends that competitive pressure should trigger **strategic evaluation**, not automatic imitation.

5.6 The scaling challenge

The evidence suggests that moving from experimentation to scale can introduce additional challenges.

These include:

- increased computing requirements;
- greater financial commitment;
- data requirements;
- employee capability;
- integration;
- governance;
- maintenance.

The MBRSG evidence is particularly useful here because it identifies advanced resources, financing, governance, and scaling considerations within the UAE AI/digital SME ecosystem. (MBRSG)

This means an AI pilot should not only ask:

> **"Does the technology work?"**

It should also ask:

> **"Can the business afford, manage, govern, and scale it?"**

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6. UAE SME AI Adoption Framework

Based on the evidence, this study proposes an **evidence-informed seven-phase framework**.

Phase 1 — DEFINE

Identify a genuine business problem.

Questions:

- What problem exists?
- Why does it matter?
- What is the current cost or inefficiency?
- What outcome would represent improvement?

Principle:

Business problem first. AI tool second.

Phase 2 — ASSESS

Evaluate six readiness dimensions:

Leadership

Is management committed?

People

Do employees have appropriate capabilities?

Technology

Are the necessary technical resources available?

Data

Is relevant and usable data available?

Finance

Can the organization support implementation and potential scaling?

Governance

What privacy, security, reliability, accountability, and compliance considerations exist?

Phase 3 — SELECT

Choose one high-value and manageable AI use case.

Selection should consider:

Potential value + feasibility + risk

Phase 4 — PILOT

Start small.

The pilot should have:

- a defined objective;
- responsible ownership;
- a timeframe;
- a budget;
- measurable success criteria.

Phase 5 — MEASURE

Evaluate actual results.

Possible measures include:

- time saved;
- cost;
- productivity;
- error rates;
- output quality;
- customer response time;
- employee acceptance.

The appropriate indicators should depend on the use case.

Phase 6 — GOVERN

Before scaling, evaluate:

- privacy;
- security;

- reliability;
- human oversight;
- transparency;
- accountability;
- applicable regulatory requirements.

Governance should be proportional to the risk associated with the application.

Phase 7 — SCALE

Scale only when:

1. measurable value has been demonstrated;
2. resources are available;
3. risks are manageable;
4. the solution can be maintained sustainably.

Final outcome:

> **Strategic + Measurable + Responsible + Scalable AI adoption**

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7. Recommendations

7.1 Start with business problems

SMEs should identify the business problem and expected outcome before selecting an AI tool.

7.2 Establish management ownership

An owner, manager, or designated project leader should be responsible for the AI initiative.

7.3 Build AI capability progressively

SMEs should develop:

AI literacy → role-specific capability → implementation/governance capability.

7.4 Use small pilots

Pilot projects allow SMEs to test value before making larger investments.

7.5 Measure outcomes

AI usage should not itself be treated as the measure of success. SMEs should evaluate actual business outcomes.

7.6 Integrate responsible AI

Privacy, security, transparency, accountability, and human oversight should be considered from the beginning.

7.7 Use external resources strategically

Where advanced infrastructure or specialist expertise is beyond internal capabilities, SMEs can consider external AI platforms, cloud services, specialist providers, and partnerships.

7.8 Scale selectively

SMEs should scale only when the evidence supports the business case and the organization can manage the associated resources and risks.

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8. Limitations

This study has several limitations.

First, it is based on secondary evidence and therefore does not provide new primary data from UAE SME owners or employees.

Second, the strongest quantitative UAE evidence examined **hospitality SMEs**, so its statistical findings should not be generalized automatically to all UAE SMEs.

Third, the MBRSG research focuses on **AI and digital SMEs**, which may be more technologically advanced than traditional SMEs.

Fourth, international literature was used to provide broader context, but international findings cannot automatically be treated as UAE-specific evidence.

Finally, the proposed seven-phase framework is **evidence-informed but not empirically validated**. Future research should test it through cross-sector UAE SME surveys and interviews.

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9. Future Research

Future studies could strengthen this research by:

1. Surveying UAE SMEs across multiple sectors.
2. Comparing AI adoption between micro, small, and medium-sized firms.
3. Conducting interviews with SME owners and managers.
4. Measuring actual AI implementation outcomes rather than adoption intention alone.
5. Testing the proposed seven-phase framework empirically.
6. Examining differences between industries such as retail, hospitality, logistics, finance, healthcare, and professional services.
7. Studying how UAE SMEs manage AI-related privacy, security, and governance risks.

Such research would allow the proposed framework to move from an evidence-informed conceptual model toward an empirically tested model.

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10. Conclusion

AI is increasingly relevant to SMEs in the UAE, but effective adoption is not simply a question of whether an SME can access AI technology.

The evidence examined in this study indicates that technological, organizational, and environmental factors interact in shaping AI adoption. UAE hospitality SME research identifies significant relationships involving top-management support, competitive pressure, government regulation, perceived usefulness, competitive advantage, and employee capability. UAE AI ecosystem research additionally highlights advanced technical resources, financing, governance, talent, compliance, and scaling considerations.

The broader SME literature reinforces the importance of skills, data readiness, implementation costs, complementary investment, and uncertainty about returns.

Taken together, the evidence suggests that the central challenge is increasingly one of **organizational capability and effective implementation rather than simple technological access**.

To address this challenge, this study proposes an evidence-informed seven-phase framework:

> **Define → Assess → Select → Pilot → Measure → Govern → Scale**

The framework does not claim to guarantee successful AI implementation. Instead, it provides a practical decision pathway through which SMEs can connect AI investments to genuine business problems, evaluate their readiness, test solutions on a manageable scale, measure outcomes, address responsible-AI considerations, and scale only when justified.

The central conclusion is therefore:

For UAE SMEs, the question is not simply whether AI can be adopted, but whether it can be adopted in a way that creates measurable business value while remaining responsible, manageable, and scalable.

References

Alkhoudi, S., Shaer, S., & Salem, F. (2025). *The artificial intelligence SMEs ecosystem in the UAE: Overcoming challenges, expanding horizons*. Mohammed Bin Rashid School of Government. (MBRSG)

OECD. (2021). *The digital transformation of SMEs*. OECD Publishing.

OECD. (2024). *Artificial intelligence and the digital transformation of SMEs*. OECD Publishing.

OECD. (2025). *AI adoption by small and medium-sized enterprises*. OECD Publishing.

Thomas, G., Albishri, N. A., Ul Islam, J., & Tanveer, M. (2025). Exploring the determinants of artificial intelligence adoption intention in the SMEs of United Arab Emirates. *SAGE Open*. (SAGE Open)

United Arab Emirates Ministry of Economy. *National Forum for SMEs / SME ecosystem information*. (Ministry of Economy)

United Arab Emirates Government. *AI policy and responsible artificial intelligence principles*. (UAE AI Office)

Artificial Intelligence Adoption in SMEs: Survey Based on TOE–DOI Framework, Primary Methodology and Challenges. (2025). *Applied Sciences, 15*(12), 6465.

Artificial intelligence adoption dynamics and knowledge in SMEs and large firms: A systematic review and bibliometric analysis. (2025). *Journal of Innovation & Knowledge*.